

ORIGINAL ARTICLE

Agricultural Soil and Food Systems

Impact of three irrigation scheduling tools on irrigation and alfalfa productivity under center pivots

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Abstract

Determining the amount and timing of irrigation events using scientific irrigation scheduling (SIS) may help optimize water use. Soil moisture sensors, commercial irrigation schedulers, and water balance programs are common SIS tools. These three methods were evaluated to test their impact on alfalfa (*Medicago sativa*) mass, nutritive value, and irrigation productivity, in comparison to experience-based irrigation depths chosen by cooperating growers. Trials were conducted at 10 farms across central Utah in 2019. Trials were repeated at nine of these farms in 2020 and six in 2021. Alfalfa mass was measured in a total of 47 cuttings from across all these fields over 3 years. The three SIS methods only impacted alfalfa mass in five cuttings, and it occurred inconsistently at various fields and years. Three cuttings had improved mass with SIS methods and two had reduced production. Forage nutritive value was more often impacted by SIS method than mass, but impacts were rarely large enough to change forage market value. Applied water was lower with most SIS methods than the grower control in 2019 and 2021 but not 2020. This was influenced heavily by the drought conditions and water restrictions during the latter 2 years of this study. As one of the first studies to directly compare how four irrigation scheduling methods for center pivots affect crop production and irrigation levels, results indicated that all three SIS approaches had comparable performance, and in some situations (especially wet years) could reduce applied water by 6%–25% without impacting alfalfa mass or nutritive value.

Plain Language Summary

This study examined how different methods of planning irrigation can help farmers use water more efficiently while still maintaining good crop yields, especially during times of drought or limited water supply. The methods tested included soil moisture sensors, commercial irrigation schedulers, and water balance programs. These were compared to irrigation schedules set by experienced farmers. The study took place on 10 farms in central Utah over 3 years, from 2019 to 2021. Researchers measured

how much alfalfa (a type of forage) was produced and the nutritive value of that forage. They found that the different irrigation methods had only small and inconsistent effects on the mass of alfalfa. However, the quality of the forage (when fed to livestock) was more often affected, though these changes were usually not large enough to impact its market value. One key finding of the study was that these new irrigation methods helped use less water compared to the methods chosen by the farmers, especially in 2019 and 2021, when there were drought conditions and water restrictions. In some cases, the methods reduced water use by up to 25% without harming alfalfa production. This was particularly noticeable in 2019 when the irrigation methods helped account for rainfall better than the farmers' traditional schedules. Overall, the study showed that these methods could be useful tools for saving water while maintaining good crop production.

1 | INTRODUCTION

In the United States, 54% of all crop sales come from irrigated land that comprises less than 20% of the harvested cropland (USDA-ERS, 2022), signifying how important irrigation is to meeting the needs of Americans. Another facet is that about 2335 people are added to the US population each day (US Census Bureau, 2022), furthering the demand for commodities grown on what has been a diminishing number of hectares nationally (FAO, 2011). The average farmland loss from 2014 to 2021 was nearly 690,000 ha year⁻¹ according to the United States Census of Agriculture (USDA-NASS, 2022).

Irrigation has proven to be a tool that can increase crop yields (Kukul & Irmak, 2019) and will likely be increasingly important in meeting the current and future population demands. Methods for optimizing irrigation use in agriculture are needed to ensure that the high demands of those limited irrigated areas can still be met. One possible solution is using scientific irrigation scheduling (SIS), including soil moisture sensors, commercial irrigation schedulers, or water balance programs to more precisely schedule irrigation amounts and timing (Gu et al., 2020).

At a conference of the American Society of Agricultural Engineers, Howell (1996) presented the limitations of irrigation scheduling, suggesting that it could not save significant amounts of water by reducing transpiration because it would reduce yield. Since that conference, significant advancements have been made in regulated deficit irrigation and reducing irrigation at specific periods in crop growth cycles has been shown to reduce water use and increase water use efficiency with minimal or no impact on crop yield and quality (Chai et al., 2016; Lindenmayer et al., 2011).

High consumer demand and low margins received by growers requires irrigation management solutions that maintain yield, eliminate wasteful application, and do not affect transpiration. A good starting point for this may be accurately

evaluating what level of water is available to the plant roots. This often happens on the farm with shovels or soil probes, but sensors can also provide reliable, useful information for knowing and responding to irrigation needs (Campbell & Campbell, 1982). In a large study in Nebraska (Kranz et al., 1992), 32 corn (*Zea mays*) fields with no irrigation scheduling method implemented were compared to 19 corn fields using tensiometers at 300-, 600-, and 900-mm depths at three to four places in the field. They (Kranz et al., 1992) found that 11% less water was applied in the fields utilizing tensiometers, and yield was maintained or increased. This would suggest that the tensiometers as a component of irrigation scheduling helped reduce overirrigation, without reducing crop transpiration; however, the study did not include replicated treatments within the same field. Challenges to soil moisture sensors that may limit their effectiveness or feasibility on some farms include inaccuracies from incorrect calibration (Sui, 2018) or difficult installations that require minimal soil disturbance (Aguilar et al., 2015). Another challenge is soil variation that may require many sensors throughout a field to attain an accurate field representation, while placing the sensors in positions representative of the root zone poses another challenge (Jones, 2004). Equipment and labor costs of installing sensors at several places in a field can quickly become expensive, particularly if the grower is not guaranteed a monetary return on the investment. There is also a degree of inconvenience from having to modify field operations to avoid damaging sensors and loggers.

One commercially available tool for irrigation scheduling is FieldNET Advisor (FNA) from Lindsay Corporation. Using soil maps and water storage data provided by the Natural Resources Conservation Service, proprietary crop canopy and root growth models, and local weather data, FNA estimates available soil water in each soil type in a field. From that data, it creates a variable rate irrigation plan that can alter the speed of the pivot to apply differing amounts or depths of irrigation.

In a study conducted by Lindsay Corporation (2018), FNA increased corn yields by 3% while applying 17% less water. There is little, if any, published, third-party research of FNA.

Several universities and private entities have developed irrigation scheduling programs or applications that use weather data from local stations, crop growth models, and user input to determine a root zone water balance for scheduling irrigation. These programs are usually simple to use, do not require equipment to be installed in the field, and can be inexpensive or free. Brajovic et al. (2015) provided an overview of some of these programs and noted that they are important tools to help feed the growing population. Vellidis et al. (2016) created an irrigation scheduling tool for cotton (*Gossypium hirsutum* L.) and found that using it increased plant-available water and water use efficiency. However, that same study (Vellidis et al., 2016) also stated two challenges to using an irrigation scheduling program, noting that precipitation at the field can be much different than the precipitation measured at the local weather station, and tillage impacts not accounted for by the programs introduce inaccuracies in the irrigation recommendations. Peters (2014) at Washington State University developed a free irrigation program called Irrigation Scheduler Mobile (ISM) that is available across the western United States and Canada. Additional coverage is possible by adding private weather stations to their network (Peters, 2014). The ISM uses local weather data, a crop coefficient, and soil available water capacity estimates based off soil texture to determine available water content of the root zone. It then displays these data in a fuel-gauge style graphic, with “Full” being field capacity (FC) and “Dead” being the permanent wilting point, as well as a read-out with the current estimated soil water deficit and time until crop water stress. Once a field is set up, the only information required by the grower is to enter irrigation amounts and harvest dates (if irrigation occurs after a harvest as is the case with alfalfa) after they occur so the ISM is properly calibrated for forthcoming irrigation recommendations. There are also options for users to fine-tune crop coefficients, soil available water holding capacity, crop growth, and actual precipitation at the field to increase the accuracy of the recommendations. The ability to make adjustments in ISM may help it overcome some of the cited inaccuracies of irrigation scheduling tools and may help it to be a useful tool for tracking soil available water and knowing the precise needs of the crop.

The objective of this study was to evaluate the effects of four irrigation scheduling approaches (cooperating grower control, a water content soil moisture sensor set, FNA, and the ISM) on alfalfa harvest mass, forage nutritive value, and water use. This is one of the first studies comparing four irrigation scheduling methods for center pivot systems and how they affect crop production and water use. Determining the reliability and possible improvements from these approaches can have substantial impacts on irrigated agriculture, particu-

Core Ideas

- Three irrigation scheduling tools were tested using 27 site-years from 47 total alfalfa cuttings in Utah.
- The three tools impacted alfalfa mass in only five cuttings at random fields and years.
- Forage nutritive value is often impacted by irrigation tools, but improvements were often not economic.
- Applied water was lower with the three tools than the grower control in 2019 and 2021 but not 2020.
- All three tools had similar performance and sometimes reduced applied water by 6%–25% without impacting production.

larly as farms grow larger and new generations begin making management decisions.

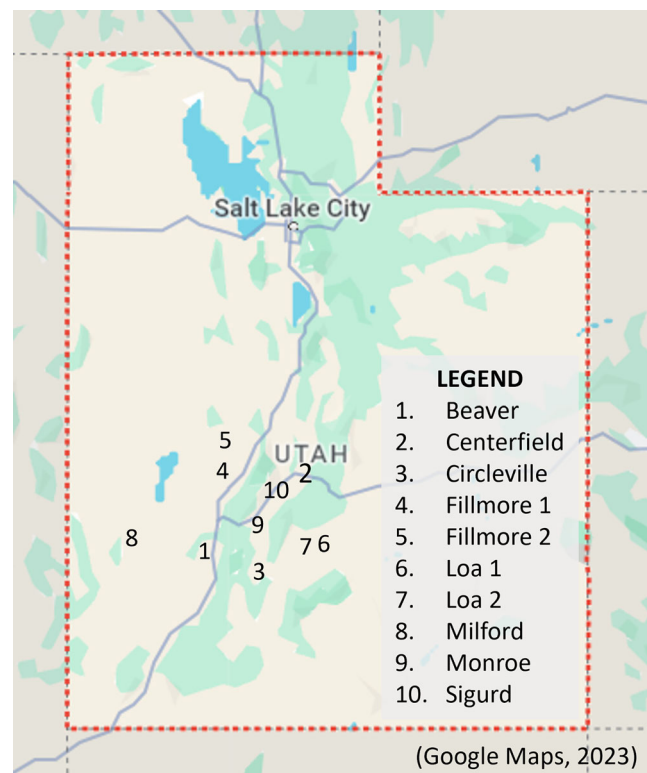
2 | MATERIALS AND METHODS

2.1 | Site characteristics

This experiment was established in the spring of 2019 on 10 farm fields in central Utah, prior to the start of the growing season (Figure 1). Due to various logistical and equipment constraints, the study was continued at nine of the 10 fields in 2020 and six for the final season of 2021. Changes in land ownership, mechanical problems with pivots, and complications with collaborating growers were the factors that determined when a field was removed from the study. All fields were established alfalfa stands with center pivot irrigation. Characteristics of the center pivots and sprinklers used in the study are included in Table 2. Soil classification, textural group, and slope classifications from the USDA-NRCS soil survey were obtained from the University of California-Davis SoilWeb application (O’Geen, 2020; Table 1). Weather data were obtained from the Utah Climate Center (climate.usu.edu; Logan, UT; Figure 2) and included cumulative precipitation and growing degree days (GDD). Weather stations were located on the field edge or within 5 km of each field. Fields with close proximity to existing weather stations were prioritized in field selections. The GDD were calculated as the daily average of the minimum and maximum air temperature minus the base air temperature of 5°C and an adjusted maximum air temperature of 30°C. Measured cumulative precipitation and GDD were compared with their respective 30-year normals (1991–2020) provided by the National Oceanic and Atmospheric Administration (noaa.gov; Silver Spring, MD, accessed December 12, 2024).

TABLE 1 Site properties from the soil web survey for irrigation scheduling tool trials in Utah from 2019 to 2020.

Site	Year	Coordinates	Dominant soil texture (classification)	Slope (%)
Beaver	2019–2021	38.305, –112.658	Loam (Calcic Argixerolls, Fine-loamy over sandy or sandy-skeletal, mixed, mesic Calcic Argixerolls)	1–3
Centerfield	2019–2021	39.100, –111.800	Silt loam (Typic Natrargids, fine-silty, mixed, Mesic)	1–2
Circleville	2019–2021	38.164, –112.265	Loam (Fine-loamy, mixed, superactive, mesic Cumulic Haploxerolls)	0–2
Fillmore 1	2019–2021	38.949, –112.433	Silt loam (Xeric Haplocalcids, fine-loamy, carbonatic, Mesic)	0–2
Fillmore 2	2019–2020	39.037, –112.419	Silt loam (Fine-silty, mixed, mesic Torrifluventic Haploxerolls)	N/A ^a
Loa 1	2019–2020	38.382, –111.583	Silt loam (Fine-silty, mixed, superactive, frigid Ustic Haplocambids)	0–3
Loa 2	2019	38.369, –111.635	Loam (coarse loamy, mixed, superactive, frigid Ustic Haplocalcids)	2–8
Milford	2019–2020	38.485, –113.421	Sandy clay loam (Fine-loamy, mixed, superactive, mesic Typic Haplocalcids)	0–2
Monroe	2019–2021	38.629, –112.157	Sandy loam (Coarse-loamy, mixed, superactive, mesic Xeric Haplocalcids)	2–4
Sigurd	2019–2021	38.839, –111.978	Loam (Coarse-loamy, carbonatic, mesic Xeric Torrifluvents)	0–4

^aN/A means not available.**FIGURE 1** Map of Utah depicting the location of the farms in the irrigation scheduling study and the name of the nearest city.

At each farm, field operations were conducted by the cooperating grower, and irrigation management was a cooperative effort among collaborators. The cooperating growers also

determined when to harvest each of their alfalfa cuttings. The source of irrigation water was surface water in all fields with a mixture of on-demand and rotation-based irrigation availability.

2.2 | Irrigation scheduling approaches

The three irrigation scheduling tools were compared with each cooperating grower's chosen application amount. To accomplish this, the cooperating growers would communicate their desired timing and amount for each irrigation as a depth. Given that this was an on-farm study, it was not possible to adjust the timing of irrigation with the three irrigation scheduling tools because they were randomized in small sectors. Therefore, only the irrigation depths were compared among the four treatments within the constraints of the grower's irrigation schedule and water availability. None of the cooperating growers used SIS methods to determine their application amounts or timing but based decisions primarily off field knowledge, past experience, historical irrigation amounts, and water availability. Restrictions in water supply did occur at many sites in 2020 and 2021 due to a widespread drought (NOAA, 2022). All treatments operated within the constraints of water allotments. The number of irrigations varied by site and year with a range of one to five irrigations and 35–260 mm applied per alfalfa cutting. Using speed control in FieldNET, application amounts from the three SIS tools and the cooperating grower's desired amount were each applied

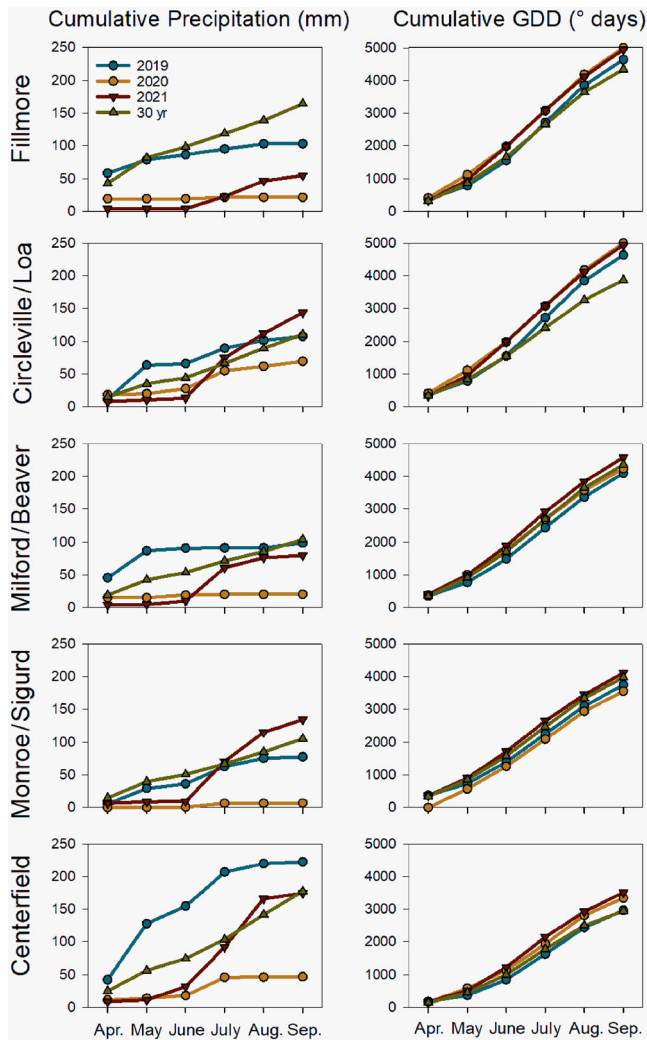


FIGURE 2 Cumulative precipitation and growing degree days (GDD) for 10 on-farm trials in Utah, gathered from the closest public weather stations to each site (some had the same station). The GDD were calculated from the minimum and maximum air temperatures and adjusted to 5°C and 30°C. The 30-year normal (1991–2020) is shown for reference.

to a large section of the field in 2019 (12°–16° each). For 2020–2021, the design was changed to include four, smaller sectors (3°–5° each) for each treatment in the field, for a total of 16 sectors assigned randomly (Figure 3). This change was implemented to improve randomization and replication of the irrigation scheduling approaches. The rest of the field was then irrigated to the cooperating grower's desired irrigation depth. This sequence was followed for every irrigation event at each field unless a periodic uniform fertigation or chemigation application was needed. These were typically low irrigation application depths to rapidly apply chemicals through irrigation. Harvest areas for forage mass measurements were taken from the middle of each sector, in a plot area that measured 3 m × 5 m and was the same area for each alfalfa harvest. Irrigation application depths were recorded for each

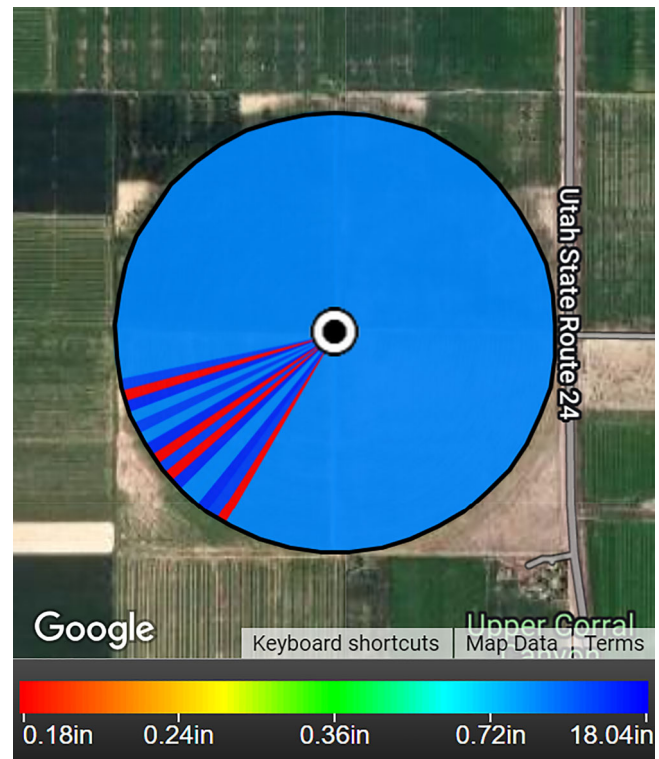


FIGURE 3 Screenshot from FieldNet Advisor in 2020 of the 16 irrigation scheduling treatment plan (four replications of four scheduling approaches [grower control, soil moisture sensor, commercial scheduler (FieldNet Advisor), and Irrigation Scheduler Mobile]) implemented in 2020 and 2021 in the lower left quadrant of one of the 10 fields in central Utah. In this prescription, irrigation application depths ranged from 0.18 to 3.0 in. (5–76 mm). The other nine fields had similar designs.

of the treatments in FieldNet software. Depths were compared among treatments and were also used to calculate an irrigation productivity index, which was the total dry matter crop yield across cuttings divided by total irrigation depth across cuttings. The index was calculated only for alfalfa cuttings with measured mass data.

2.3 | Soil moisture sensor with a water balance spreadsheet

Prior to the 2019 growing season, a single soil moisture sensing station was installed in each field. Each station had six of the Teros 10 volumetric water content (VWC) sensors (Meter Group) installed at 300-, 600-, 900-, 1200-, 1500-, and 1800-mm depths by insertion into the wall of a 100-mm bore hole. At three farms (Beaver, Fillmore 1, and Milford), sensors were installed at 150-, 300-, 450-, 600-, 750-, and 900-mm depths because the equipment to install deeper was not yet available. Sensor stations were installed within one of the four replications of the soil sensor treatment in the same plot

area where forage mass and nutritive quality samples were collected (to be described). Soil moisture data were collected by ZL6 data loggers (Meter Group) monitored remotely through Meter Group's ZENTRA Cloud online application. The sensors were not calibrated beyond factory calibration settings from Meter Group. The wet spring of 2019 (Figure 2) was ideal for estimating FC at each sensor depth. After significant rain events, the water content was monitored closely to observe when the soil would quickly drain excess water, and then pause at a constant level. This point was recorded as the estimated FC for each soil moisture sensor.

To generate irrigation recommendations from the VWC readings, a simple spreadsheet was created where the current VWC readings were subtracted from the previously determined FC at each depth down to 900 mm. Those calculations were summed together to determine the amount of irrigation required to refill the root zone. That value was then divided by 80% to account for pivot efficiency losses common to the mid-elevation sprinkler application (MESA) sprinkler packages used on the pivots in the study (Amosson et al., 2011; New & Fipps, 2000; Sarwar et al., 2019). On sites with sensors below 900 mm, soil moisture was observed at the deeper depths, but the effective root zone for irrigation recommendations was considered to be 900 mm (Alam & Rogers, 2009; Irmak et al., 2007). Though this was not commonly an issue, irrigation amounts were never permitted to exceed 90 mm in a single irrigation event. A management allowable depletion level was set at 60% as a guide to keep VWC above that level to limit crop stress. This was followed as closely as water schedules and water availability would permit. The VWC in the one replication was used to set a single irrigation depth for all four replications of this SIS treatment due to economic restrictions of installing four soil moisture stations at each site to represent each replicate. Annual costs for this irrigation scheduling method are estimated at around \$2000 per pivot, according to the Utah State University Extension Irrigation Technology Cost/Benefit Analysis calculator (USU, 2023), when installing a single soil moisture sensing station. This cost spreads the upfront costs of equipment and installation over 5 years (conservative lifespan of equipment), which accounts for annual subscriptions and operational labor.

2.4 | Commercial irrigation scheduler

The computer program used to apply the irrigation treatments was FieldNET from Lindsay Corporation. An option within FieldNET is to add additional programming called FNA. According to their product summary, FNA can "track the available soil water throughout the field by combining a soil map of the field, proprietary dynamic crop canopy and root growth models, hyper-local weather data, and the applied irrigation history" to generate irrigation recommendations.

The program divides a circular field into 360 sectors, that each receive a recommendation from the data sources previously mentioned. According to 2023 pricing, the costs associated with this tool are an annual FieldNET subscription (~\$255 per pivot), an annual FNA subscription (~\$383 per pivot), and the labor to manage the app. Farms with several pivots can benefit from small price breaks on the subscriptions or a reduced annual FieldNET subscription of \$153 per pivot if a radio base station (~\$2000 for up to 100 pivots) is utilized with on-farm internet (R. Moyle, personal communication, February 1, 2023). To test FNA as the example of a commercial irrigation scheduler, the sectors that contained the plot area would receive the amount of irrigation as recommended by the program on the day the cooperating grower chose to irrigate. For the 2019 and 2020 growing seasons, this treatment was applied for each individual degree within the plot areas. In 2021, recommendations were not entered correctly for these treatment areas, as the recommendation for the entire field minus the other treatment areas was mistakenly used instead of the individual degree prescriptions. Therefore, results from this year and treatment need to be considered in this context.

2.5 | Irrigation scheduler mobile

Each field was entered into ISM (Peters, 2014) prior to the start of the 2019 growing season. Throughout the growing seasons, ISM estimates the available water content of the soil and displays what the current soil water deficit is. Like the procedure for using a soil moisture sensor, the deficit was divided by 80% to account for irrigation water lost due to efficiency losses of the pivot sprinkler package. Every irrigation event was recorded in ISM to keep the water balance up to date. Harvest dates of the alfalfa were also input into ISM to keep the model accurate for changes in evapotranspiration as the crop is harvested and begins to regrow. Though the program is free, it is estimated that \$824 per pivot is required annually to cover the labor costs of operating the program over a 20-week irrigation season (USU, 2023).

2.6 | Crop mass and nutritive value

Alfalfa harvest mass was measured after cooperating growers had windrowed each field, but before the windrows were raked. Growers harvested according to their preferred schedules using windrowers that were 5–6 m wide. In the region where the study occurred, most growers harvest at least three cuttings of alfalfa each year and some can harvest four when adequate irrigation is available. The study was established at the beginning of the growing season in 2019 and resulted in harvest data from the second, third, and fourth cuttings in 2019. In 2020, all eight sites except Sigurd had three cuttings

but the third cutting data were lost at the eight sites due to a widespread high-speed wind event that severely scattered and mixed cut hay. In 2021, severe drought conditions limited water supply on some farms resulting in a single cutting of data at Centerfield, Circleville, and Sigurd. Three other sites (Beaver, Fillmore 1, and Monroe) had more available water and received their regular three to four cuttings.

At all sites, a length of 3 m was measured from each windrow and all alfalfa in that section was collected and weighed within each of the 16 plot areas in each field. This resulted in four replications per treatment. The plots were positioned in the same pivot span that had the soil moisture sensor stations (Table 2). A subsample (about 100 g) was weighed in the field and then forced-air oven dried at 60°C until constant mass and reweighed to measure moisture and calculate dry matter mass of each plot. Subsamples were ground to pass through a 1-mm sieve using a Thomas Model 4 Wiley Mill. Ground samples were analyzed by near-infrared reflectance spectroscopy (NIRS) using a FOSS DS2500 F (Foss North America Inc.) and the 2019–2021 legume hay NIRS consortium equations (NIRS Forage and Feed Consortium, Berea, KY) to estimate dry matter, ash, fat, crude protein (CP), acid detergent fiber (ADF), neutral detergent fiber (NDF), and neutral detergent fiber digestibility 48 h (NDFD). The following two quality parameters were calculated:

Total digestible nutrients (TDN)

$$= (100 - (\text{NDF} - 2 + \text{CP} + 2.5 + \text{Ash})) \times 0.98 + \text{CP} \times 0.93 + (\text{Fat} - 1) \times 0.97 \times 2.25 + (\text{NDF} - 2) \times \text{NDFD}/100 - 7, \quad (1)$$

Relative forage quality (RFQ)

$$= (((0.012 \times 1350 / (\text{NDF}/100) + (\text{NDFD} - 45) \times 0.374) / 1350) \times 100) \times \text{TDN}/1.23. \quad (2)$$

2.7 | Statistical analysis

Data from each individual harvest at each farm were analyzed for differences among the dependent variables of crop mass and the nutritive value parameters (Table 3). Treatment was considered a fixed effect and no interactions of year or site were evaluated due to design changes between years, and communication errors with cooperating growers and weather events that resulted in missed harvests. Differences were considered significant at $p < 0.05$ while using the MIXED procedure of SAS (SAS Institute Incorporated version 9.4). The UNIVARIATE procedure of SAS was used to

TABLE 2 Center pivot and sprinkler properties for on-farm irrigation scheduling tool trials in Utah from 2019 to 2020.

Site	Total target flow rate (L min^{-1})	Top of pivot pressure (Kpa)	Length (m)	Towers	Sprinkler count	Sprinkler brand (type)	Regulator pressure (Kpa)	Span with plots	Area irrigated (ha)
Beaver	3222	206.8	408.8	7	107	Nelson (R3000)	68.9	6	49.8
Centerfield	4548	317.2	550.9	10	176	Nelson (S3000)	68.9	9	102.5
Circleville	1819	165.5	406.3	7	150	Senninger (I-wob)	68.9	6	52.7
Fillmore 1	758	137.9	203.1	4	72	Nelson (D3030)	103.4	3	15.4
Fillmore 2	3411	186.2	350.0	6	107	Senninger (I-wob)	103.4	4	42.9
Loa 1	3411	296.5	391.6	6	140	Senninger (I-wob)	68.9	5	53.9
Loa 2	2653	275.8	365.0	6	107	Nelson (R3000)	137.9	4	39.7
Milford	4264	275.8	423.4	7	114	Nelson (R3000)	—	6	53.9
Monroe	4738	413.7	400.0	6	145	Senninger (I-wob)	68.9	4	47.8
Sigurd	1364	165.5	227.8	4	71	Senninger (I-wob)	103.4	3	19.4

TABLE 3 Significant ($p < 0.05$) differences of F -tests for the fixed effect of scheduling method on alfalfa mass and nutritive value parameters from 47 measured cuttings during 2019–2021 in central Utah.

Response variable	Year	Site	Cutting	$P > F$
Mass	2019	Sigurd	3	0.014
		Sigurd	4	0.025
	2021	Sigurd	1	0.001
		Fillmore 1	2	0.036
		Monroe	1	0.050
Crude protein	2019	Beaver	3	0.004
	2020	Milford	1	0.019
Neutral detergent fiber	2019	Beaver	3	0.000
		Milford	3	0.022
	2020	Sigurd	3	0.041
		Sigurd	4	0.042
		Centerfield	1	0.013
		Milford	1	0.033
Neutral detergent fiber digestibility	2019	Fillmore 1	3	0.035
		Fillmore 1	4	0.026
		Monroe	3	0.031
		Fillmore 2	3	0.002
		Fillmore 2	4	0.004
	2020	Sigurd	4	0.033
		Centerfield	1	0.002
		Centerfield	2	0.018
	2021	Sigurd	1	0.002
		Fillmore 1	2	0.024
Total digestible nutrients	2019	Beaver	3	0.000
		Loa 2	2	0.038
		Loa 1	3	0.047
		Fillmore 2	3	0.034
	2020	Centerfield	1	0.012
	2021	Beaver	1	0.041
		Sigurd	1	0.004
Relative forage quality	2019	Beaver	3	0.000
		Milford	3	0.033
	2020	Sigurd	3	0.030
		Circleville	1	0.036
		Centerfield	1	0.005
		Milford	1	0.005
	2021	Sigurd	1	0.041

Note: Only significant differences are shown for brevity.

inspect residuals to ensure that the assumptions of normality and equal variance were satisfied. All mean separations were conducted using Fisher's protected least significant difference at $\alpha = 0.05$ using the PDIFF option in the MIXED procedure of SAS.

3 | RESULTS AND DISCUSSION

Upon completion of the study, researchers learned that net irrigation amounts were supposed to be entered into ISM, not gross irrigation amounts. For instance, if ISM reported a deficit from FC of 50 mm, a 50-mm irrigation depth was programmed into the pivot and applied, then entered into ISM. The correct procedure would have been to calculate $50 \text{ mm} \div 80\%$ (efficiency) and use the resultant 63 mm as the amount programmed into the pivot, while only reporting 50 mm in ISM. Due to this oversight, the water balance equation used by the program would have been incorrect by the amount of irrigation loss during application, which for MESA-style sprinklers has been measured to be about 20% (Amosson et al., 2011; New & Fipps, 2000; Sarwar et al., 2019). This error resulted in a leaner test of ISM, potentially allowing underirrigation. However, it provides unique insight into the accuracy and practicality of ISM as it is likely that many users make the same error. Thus, testing it in the fashion that we did provides an assessment of how ISM may perform in practical settings.

3.1 | Soil moisture sensors

The use of a single set of soil moisture sensors and a water balance spreadsheet most often resulted in the lowest irrigation application depth of all four scheduling methods. In the 20 site-years where irrigation depths could be compared, the soil moisture sensor treatment had recommended irrigation depths equal to (i.e., within 25 mm) the control at four site-years or less than (>25 mm reduction) at eight site-years (Table 4). At the eight site-years where sensors reduced irrigation depths, the average decrease in irrigation was 215 mm. These reductions occurred at three sites in 2019, two in 2020, and three in 2021 with 25%, 9%, and 23% reductions in application depths compared to the control, respectively. When averaged across sites each year, the sensor treatment recommended 20 mm less, 49 mm more, and 100 mm less than the control in 2019, 2020, and 2021, respectively. These results indicate that the soil moisture sensor consistently recommended less irrigation at some sites in all 3 years, with the largest reduction in the wettest year (Figure 2).

Out of 47 measured harvests, the soil moisture sensor treatment area never had significantly reduced alfalfa mass compared to the control, and it increased mass twice at the Sigurd site, from 4.0 to 4.6 Mg ha^{-1} (15%) in 2019 and from 2.1 to 3.3 Mg ha^{-1} (57%) in 2021 (Table 3; Figure 4). The soil moisture sensor treatment affected CP levels twice (Table 3). Once was in 2019 where the sensor produced alfalfa with 28.9% CP compared to 27.6% for the ISM treatment and 25.9% for the FNA treatment. The other case was in 2020, when the sensor treatment reduced CP levels to 23.8%

TABLE 4 Total irrigation depths (mm) summed across harvested alfalfa cuttings by year and site for three irrigation scheduling treatments (soil moisture sensors [SMS], commercial scheduler [FieldNet advisor; FNA], and irrigation scheduler mobile [ISM]) and a grower control tested at 10 sites in central Utah.

Year	Treatment	Beaver	Centerville	Circleville	Fillmore 1	Fillmore 2	Loa 1	Loa 2	Milford	Monroe	Sigurd
2019	Control	859	881	324	.	.	.	85	643	290	542
	SMS	625	928	.	.	.	.	115	460	423	435
	FNA	785	828	519	.	.	.	148	422	401	479
	ISM	770	840	432	.	.	.	229	594	562	509
2020	Control	275	724	.	768	1003	N/A	331	439	1019	825
	SMS	338	816	.	725	1022	N/A	327	388	1018	1141
	FNA	535	779	.	1101	1106	N/A	754	509	1726	1179
	ISM	442	695	.	969	932	N/A	655	442	1299	1059
2021	Control	880	311	210	2452	N/A	N/A	N/A	N/A	888	259
	SMS	1163	331	173	1496	N/A	N/A	N/A	N/A	779	461
	FNA	1133	349	432	1876	N/A	N/A	N/A	N/A	1027	468
	ISM	951	330	373	1380	N/A	N/A	N/A	N/A	877	432

Note: N/A indicates sites where harvest data were not available, and periods indicate missing irrigation depth data. Bold values indicate cases where a scheduling tool decreased irrigation depth more than 25 mm compared to the control.

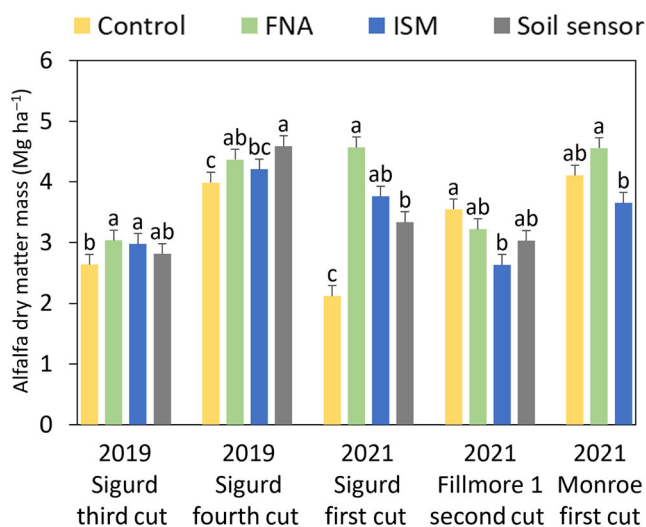


FIGURE 4 Dry matter (DM) alfalfa mass results for the five of 47 measured harvests in central Utah where mass was significantly ($p < 0.05$) impacted by scheduling method (FNA, FieldNet Advisor; ISM, Irrigation Scheduler Mobile). Mean separations were conducted at $p < 0.05$ by harvest, and differences within site-year are denoted by letters above each bar. The data for the soil moisture sensor treatment at the Monroe site are missing for the first cutting due to communication errors with the cooperating farmer.

compared to 25.2% for the control treatment (i.e., irrigation depths used by cooperating growers). All these CP levels were within the “supreme” forage quality ranking for alfalfa (USDA-AMS, 2023) and may not alter the market value of the forage, but for operations feeding their own livestock, these differences may be important, as supplemental protein in rations is an additional cost. The NDF levels are important

to understanding the digestibility of feedstuffs; high values suggest difficult digestion, and desirable low values signal a low fiber content (Caddell et al., 2001). Over the duration of the study, the soil moisture treatment raised NDF levels once above the control, from 27.5% to 32.4%, and never caused a significant decrease from the control treatment (Table 3). The NDFD (48 h) levels were reduced by the soil moisture sensor treatment in three cuttings to 52.0%, 47.0%, and 52.9%, which were reductions of 12%, 11%, and 9%, respectively, compared to the control. The sensor treatment also raised NDFD levels once by 8%, when the control treatment measured 45.6% and the soil moisture sensor treatment measured 49.1%. The TDN levels were affected by the sensor treatment three times during the study, lowering TDN levels from 76.2% to 74.0% and from 77.4% to 73.8% and raising them once from 54.5% to 60.7%, respectively, compared to the control. The RFQ calculations are a key metric for valuing forages as they estimate intake and energy and include the TDN in the calculation. Elevated RFQ is desirable and improves alfalfa market value above certain thresholds. This metric was impacted in only two cuttings by the soil moisture sensor treatment, where in both instances, it resulted in significantly lower RFQ than the control (Table 3). This occurred at a late-season cutting at the Milford site in 2019, where the RFQ was reduced by 17%, and in an early-season cutting in 2021 at the Sigurd site, where it reduced RFQ from by 12% (Figure 5).

While the soil moisture sensor had among the least amount of irrigation applied and rarely impacted alfalfa harvest mass, it did not frequently maximize irrigation productivity. The soil moisture sensor treatment had the maximum irrigation productivity among the four scheduling treatments in only five of the 18 site-years (28%) where it could be evaluated (Table 5).

TABLE 5 Irrigation productivity (IP) calculated as dry matter alfalfa yield divided by irrigation depth across all alfalfa cuttings with mass data for the grower control and three irrigation scheduling approaches in on-farm trials in central Utah in 2019–2021.

Site	Year	Grower control (kg ha ⁻¹ mm ⁻¹)	Soil moisture sensor (kg ha ⁻¹ mm ⁻¹)	Commercial scheduler (kg ha ⁻¹ mm ⁻¹)	Irrigation scheduler mobile (kg ha ⁻¹ mm ⁻¹)
Beaver	2019	8.1	10.2	9.8	9.4
	2020	8.9	5.4	5.0	6.5
	2021	9.3	7.8	8.0	10.2
Centerfield	2019	5.7	5.6	5.5	5.3
	2020	11.0	9.6	10.6	12.1
	2021	9.4	7.6	7.1	9.8
Circleville	2019	12.8	N/A	7.6	8.6
	2020	N/A	N/A	N/A	N/A
	2021	16.1	N/A	10.0	9.0
Fillmore 1	2019	N/A	N/A	N/A	N/A
	2020	3.7	3.3	2.7	2.7
	2021	4.8	5.8	5.2	6.2
Fillmore 2	2019	N/A	N/A	N/A	N/A
	2020	10.4	10.3	8.3	11.2
Loa 1	2019	29.1	17.1	19.6	11.1
	2020	18.1	N/A	9.2	9.1
Loa 2	2019	N/A	N/A	N/A	N/A
Milford	2019	3.2	5.3	4.8	3.7
	2020	10.3	11.4	8.0	10.4
Monroe	2019	12.2	8.2	9.2	6.6
	2020	7.8	8.4	4.5	6.5
	2021	10.1	6.9	9.4	9.8
Sigurd	2019	12.2	17.0	15.4	14.1
	2021	13.2	10.0	10.0	10.6
	2020	8.2	7.2	9.7	8.7
Count max. IP		10	5	1	5
Percent max. IP		48%	28%	5%	24%

Note: Values in bold were the maximum irrigation productivity within each site-year. Missing values occurred in some site-years due to lack of irrigation depths due to pivot panel malfunctions (indicated as N/A). Count and percentage of site-years with the maximum (max.) IP were included for each scheduling treatment.

This frequency was about half that of the control treatment that produced the maximum in 48% of the 21 site-years where it was evaluated.

When averaged across sites each year, the irrigation productivity of the soil moisture sensor method was greatest in 2019 and least in 2021. This difference was not the same as the other three SIS treatments where irrigation productivity was sometimes greatest in 2021. It was not clear why irrigation productivity was not greater with soil moisture sensors than the control in more cases since the sensors reduced irrigation in 40% of the site-years. It could be related to several factors besides irrigation such as crop management, nutrient management, pest pressure, and weather patterns that can also influence alfalfa irrigation productivity.

Across the 47 harvests of the study, there were few differences in mass or nutritive quality due to scheduling irrigation

based on soil water content measurements from one location in each field, and there were some semi-frequent (40% of site-years) and valuable water savings (215 mm) in many site-years when using this method compared to the control. This suggests that soil moisture sensors are likely a viable tool for reducing irrigation levels and maintaining alfalfa production. One may choose to deduce that using a set of sensors is more accurate at determining the crop needs than the other methods, simply because it is using an actual measurement inside the field. By contrast, the control and the other tools tested rely heavily on soil and crop estimates, as well as precipitation measurements that can differ widely from the weather station to the field location. Some pitfalls of using soil moisture sensors were the extra labor required to install and monitor the sensors, along with maintaining a water balance spreadsheet. Other manufacturers have user-friendly interfaces that

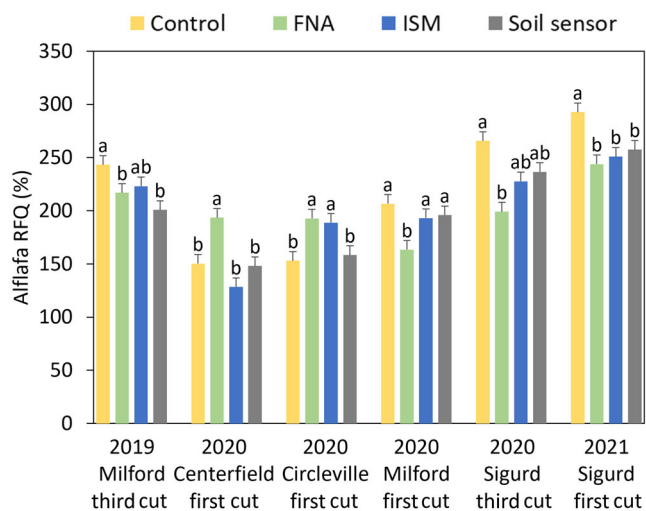


FIGURE 5 Alfalfa relative forage quality (RFQ) measurements for six of 47 measured harvests in central Utah where RFQ was significantly ($p < 0.05$) impacted by scheduling method (FNA, FieldNet Advisor; ISM, irrigation scheduler mobile). Mean separations were conducted at $p < 0.05$ by harvest, and differences by site-year are denoted by letters above each bar.

perform the calculations and generate recommendations for the user; however, the sensors used in this study were selected primarily for accuracy and telemetry. Furthermore, it can be difficult to place a single or a few soil moisture sensor stations within a field so that they adequately represent soil moisture variation in time and space (Barker et al., 2017; Zhao et al., 2018). This limitation was not addressed in this assessment of soil moisture sensors and should be considered in future research that compares scheduling tools.

Soil moisture sensors have been evaluated for irrigation scheduling in previous studies more frequently than the other two SIS approaches. Previous studies (e.g., Ganjegunte & Clark, 2017; Miller et al., 2014) have evaluated the use of soil moisture sensors for irrigation scheduling on a variety of crops and found that irrigation amounts can be reduced without impacting mass or yield. While several studies exist, most have focused on sensor accuracy (e.g., Lo et al., 2020) or theoretical applications of sensors or the fusion of sensors and models (Li et al., 2019), and few have assessed operational feasibility for variable applications (Kukul et al., 2020), especially for forage production. Our results help begin to fill this need and provide evidence that soil moisture sensors have high potential to help refine alfalfa irrigation management without sacrificing mass or nutritive value.

3.2 | Commercial irrigation scheduler

The FNA program offers a computer-generated, 360 sector, speed-controlled plan that is easily implemented. In the first

year of the study, it prescribed an average of 512 mm of irrigation across all farms for only the cuttings with available data, which was about 1% less than the control treatment application of 518 mm (Table 4). In 2020 and 2021, when the cooperating growers were making cutbacks due to drought, the commercial scheduling program prescribed about 43% and 6% more than the control treatment, with an application depth that averaged 961 and 881 mm in 2020 and 2021, respectively. Irrigation levels used by cooperating growers in the control treatment were substantially lower in 2020–2021, as many growers only had water available for part of the growing season. When results were examined by site-year, the commercial model reduced irrigation depths compared to the control by an average of 197 mm in five of 21 site-years and increased depths by an average of 221 mm at the remaining 16 site-years (Table 4). The reductions mainly occurred in 2019 where four site-years had an average of 15% less irrigation than the control. The other reduction occurred at a single site-year in 2020 where excessive irrigation was applied by the grower and FNA recommended 24% less. The site-year results confirm those across years and indicate that the commercial model was frequently recommending more irrigation than the control treatments except in 2019, the wettest year of the study.

Out of the 47 measured harvests across 2019–2021, the commercial scheduling program impacted alfalfa mass in a total of three harvests, all at the Sigurd site, raising mass each time (Table 3; Figure 4). In 2019, it increased mass compared to the control in the third and fourth cuttings, representing 15% and 10% mass increases, respectively. In 2021, mass was increased 115% with FNA compared to the control. However, this increase was with 81% more water applied to the FNA plots than the control plots, as the cooperating grower was trying to conserve water due to drought.

The effects of the FNA treatment on alfalfa CP content were minimal (Table 3), and it only impacted levels once, reducing CP from 25.2% to 22.4% compared to the control. NDF levels were increased in three harvests and reduced once by FNA, each on separate farms. In 2019, NDF levels were increased from 27.5% to 30.5%, and in 2020, NDF increased from 31.9% to 37.5% and from 26.2% to 32.8% with FNA compared to the control. Further, at one farm in 2020, NDF levels were reduced from 40.7% to 34.2%. The NDFD (48 h) measurements and TDN calculations shared similar results (Table 3), where there were differences in five and three harvests, respectively, but all values remained within an acceptable range to qualify for a supreme alfalfa rating and the highest market value. The RFQ values were reduced with FNA compared to the control by 11%, 21%, 25%, and 17% at the Milford site in 2019 and 2020 and at the Sigurd site in 2020 and 2021, respectively (Table 3; Figure 5). The 2021 reduction at the Sigurd site was likely in response to the control treatment area receiving 81% less water that year, resulting in

much less mature alfalfa. There were two harvests during the study that the FNA treatment raised RFQ levels, both in 2020, where it increased levels by 26% at the Centerfield site and 29% at the Circleville site (Figure 5). These would be valuable differences that would affect the market value of alfalfa if it were sold off-site. However, occurrences of reduced RFQ were infrequent over the 47 measured harvests of the study.

Minor or no differences in mass and frequent increases in irrigation depths of the commercial scheduling tool compared to the control resulted in it having lower irrigation productivity in 20 of 21 site-years where it could be evaluated (Table 5). Few, if any, other studies have evaluated the impact of commercial irrigation scheduling programs such as FNA on alfalfa or irrigation productivity. As the first published assessment of FNA's impact on alfalfa production, our results indicate that it had minor and inconsistent effects on alfalfa mass or nutritive value compared to the conventional irrigation schedules employed by the cooperating growers. Thus, improvements in FNA may be required before it can provide enough benefit to alfalfa to justify the added cost for growers to purchase, learn, and use the program.

3.3 | Irrigation scheduler mobile

In 2019, the ISM program average irrigation recommendation across farms for the cuttings with available data was 562 mm, which was an 8% increase compared to the control plots that averaged 518 mm (Table 4). In 2020, there was a larger difference between the ISM and control averages, as they recommended 812 and 673 mm, respectively. These results differed greatly from 2021, where the average ISM recommendation was 724 mm, and the average control recommendation was 834 mm. When site-year data were evaluated, the ISM treatment reduced irrigation by an average of 198 mm at seven of 21 site-years (33%). These seven reductions occurred in all 3 years with an average of 9% less irrigation than the control in 2019, 6% less at two sites in 2020, and a 44% reduction at a single site in 2021 (the site with excessive irrigation in control). The ISM also increased applications by an average of 194 mm across 11 site-years or produced similar (within 25 mm) depths as the control in the remaining three site-years. This means that ISM recommendations may frequently recommend more irrigation than growers apply; however, it could be a useful and inexpensive tool for supplying information for scheduling irrigation, particularly to irrigators less experienced than the cooperating growers of this study.

Over 47 alfalfa cuttings, mass was impacted in three cases by the ISM program recommendations (Table 3). In 2019, ISM increased mass compared to the control at the Sigurd site by 13% (Figure 4). In 2021, ISM increased mass by 78%

at the same farm, but that was with 78% more water applied to the ISM plots than the control, as the cooperating grower was attempting to stretch their limited water supply during an extreme drought. After observing the increased mass in the soil sensor, ISM, and FNA scheduling areas, this grower determined it would have been better to concentrate the limited water to the first cut, as the other methods had recommended (not intuitively, but only as programmed to minimize crop stress), rather than to try to stretch their water into a second cutting, as the saved water was not enough to promote enough regrowth for a second harvest. The grower's approach resulted in large yield losses due to the stunted first cutting followed by the failed attempt at a second cutting. The only other time that a mass difference occurred as a result of the ISM treatment was in 2021 when it reduced mass compared to the control by 26% at the Fillmore 1 site (Figure 4), but that was with 44% less water applied. At that field, the ISM season recommendation was for 1381 mm, while the control recommendation was 2453 mm, an amount that is not typically used by Utah growers.

The forage nutritive value analysis showed occasional differences due to the ISM treatment (Table 3), but these were limited and would have had little effect on the market value of the crop. The CP and NDF levels were never changed by the ISM treatment. The NDFD (48 h) had the greatest significant differences and was impacted by ISM in seven harvests throughout the study, but these changes were never large enough to change the monetary value of the alfalfa. The TDN calculations were only impacted by ISM in three harvests, but again the differences were minimal and would have had little effect on the value of the crop to be fed on-farm or sold off-site. The RFQ values were increased by 23% for the ISM treatment compared to the control in a single harvest at the Circleville site in 2020 (Table 3; Figure 5). This would have increased the feed and market value of the alfalfa, and because this occurred on the first cutting of the season, where yields are typically highest, it would have been particularly beneficial to the cooperating grower. The other instance where the ISM treatment affected RFQ was when it decreased RFQ by 14% at the Sigurd site in 2021 (Figure 5). Despite this decrease, the alfalfa would still be rated as supreme (USDA-AMS, 2023) and the change would not impact the market value.

The irrigation productivity of the ISM treatment ranged from 2.7 to 14.1 kg ha⁻¹ mm⁻¹ across site-years (Table 4). When averaged across site-years, irrigation productivity of ISM was greatest in 2021 and least in 2019. This scheduling tool had the maximum irrigation productivity among the four treatments in five of 21 site-years (24%) where it was evaluated (Table 5). This was the same frequency as the soil moisture sensor approach. Three of these occurrences were in 2021 and the others were in 2020. This aligns with the irrigation depth data (Table 4) showing that the ISM often

recommended more irrigation on average than the control in 2019 and 2020 and less in 2021.

The ISM developed at Washington State University by Peters (2014) is one of many soil water balance models available. Other examples available in the western United States with various levels of complexity include CropManage (<https://cropmanage.ucanr.edu/>; accessed June 6, 2025) and Water Irrigation Scheduler for Efficiency (<https://erams.com/catena/tools/agricultural-resources/irrigation-advisor/>; accessed June 6, 2025). While several models exist, few replicated comparisons (Taghvaeian et al., 2020) have been made to determine how these models influence crop mass, nutritive value, and irrigation productivity. Therefore, as one of the first replicated assessments in on-farm trials, our results indicate that ISM (and possibly other models) may often (52% of site-years) recommend more irrigation than what growers apply without increasing alfalfa mass or nutritive value. The excess irrigation may be even greater than we measured because we evaluated the ISM at irrigation levels lower than what is currently recommended by the tool. Therefore, further testing and development of the ISM and other models may be needed to improve predictions and refine estimates of the advantages and disadvantages of models relative to other scheduling approaches.

3.4 | Contrasting the scheduling approaches

For a grower who has decided to adopt a scheduling approach from the three SIS approaches evaluated, there are several variables that are important in choosing, such as cost, time, and the level of affinity one has for inputting data and completing calculations, none of which were evaluated in this study. Based only on measured performance in this study, using a set of soil moisture sensors with a water balance spreadsheet will likely produce the most conservative irrigation prescriptions that will still meet the water demands for alfalfa mass and nutritive value standards. Balancing forage mass and nutritive value tradeoffs is important in alfalfa and other forage crop production (Sohoulande et al., 2025). However, the minor impacts of the three SIS on nutritive value indicate that the three SIS approaches (as tested in this study) may not influence that balance.

The results also showed that the ISM program may not reduce the amount of irrigation applied as frequently as soil moisture sensors (33% vs. 40% of site-years), but it proved to be a safe and inexpensive option for maintaining mass and nutritive value. Commercial schedulers, such as FNA, often prescribed more irrigation for alfalfa than the control or the other two scheduling tools (76% compared to 40%–52% of site-years) without additional gains in mass, nutritive value, or irrigation productivity. This indicates that improved predictions and calibrations may be needed to

improve scheduling tools such as FNA that are based on combinations of crop growth models, soil maps, weather stations, and other remotely sensed data. Further, additional integration and testing of soil sensors fused with models (Li et al., 2019) should be evaluated in replicated field trials to determine their validity. Despite the disadvantage of the commercial tool in this study, it does require the least amount of effort to use because there is little maintenance and no manual data collection or calculations. These aspects may offer a competitive advantage over other options and warrant additional testing to improve ease and adoption for growers. From 47 measured harvests in this study, all scheduling methods performed well, with few changes to crop mass or nutritive value (Table 3) and some irrigation reductions, which suggests that any of the options can be useful tools. However, it was also apparent that the SIS approaches are not guaranteed to optimize water in all situations and that grower's experience over years of managing their irrigation is often resulting in maximum irrigation productivities.

4 | CONCLUSIONS

Each of the irrigation scheduling methods tested can provide useful information to assist growers with irrigation decisions. However, they often did not reduce water application in comparison with what the cooperating growers in this study were choosing to apply. This may have been due to the drought conditions present throughout the latter 2 years of this study. Under these conditions, all three scheduling methods, besides the control, lacked the incorporation of data that may have been even more crucial to the cooperating growers, such as how limited their water turn was or how quickly their irrigation district would be shutting off water, which was after only one or two cuttings at some sites. These factors often resulted in higher recommendations from the three scheduling methods tested. These recommendations would likely have increased alfalfa mass and would have been desired by the growers, but drought limitations often resulted in a more conservative recommendation from the cooperating growers.

At some of the farms in the study, the SIS methods (soil sensor, ISM, and FNA) helped maintain yield with less irrigation on occasion. However, there were few cases where there were ongoing benefits, and many farms had no improvements at all. For two of the three growing seasons evaluated, Utah was in a drought, and to some degree this impacted the grower recommendations used as the control of the study, as some cooperating growers attempted to stretch their water to more acres or an additional harvest. These results guide the notion that most growers are extra careful with their water resources, and SIS methods may not bring drastic changes in alfalfa production or water use to growers who are well in-tune with the conditions of each field. At the least, the SIS approaches

provide helpful insights into the soil, plant, and water relationship that can be considered by the veteran irrigator but can also be safely relied upon. As farms grow larger and more land becomes irrigated, SIS will be important in many situations for accurately matching irrigation to crop water needs. These tools can provide reliable recommendations that can be trusted to sufficiently irrigate crops and minimize overirrigating. As one of the first studies to directly compare how four irrigation scheduling methods for center pivots affect crop production and water use, the results indicated that the soil moisture sensor approach most frequently reduced irrigation applications compared to the other two SIS approaches (FNA and ISM). However, the advantage was slight compared to the ISM, and all three tools had comparable performance and irrigation productivity. It also indicated that in situations where water is plentiful, such as in 2019, the SIS approaches offer superior accounting for soil moisture conditions and may result in a 6%–25% reduction in seasonal irrigation.

AUTHOR CONTRIBUTIONS

Jonathan Holt: Conceptualization; data curation; formal analysis; investigation; methodology; project administration; resources; software; supervision; validation; visualization; writing—original draft; writing—review and editing. **Matt Yost:** Conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; software; supervision; validation; visualization; writing—original draft; writing—review and editing. **Jody Gale:** Conceptualization; data curation; funding acquisition; investigation; methodology; project administration; resources; supervision; writing—original draft; writing—review and editing. **Mark Nelson:** Conceptualization; data curation; funding acquisition; investigation; methodology; project administration; resources; supervision; writing—original draft; writing—review and editing. **Trent Wilde:** Conceptualization; data curation; funding acquisition; investigation; methodology; project administration; resources; writing—original draft; writing—review and editing. **Kalen Taylor:** Investigation; methodology; resources; writing—original draft; writing—review and editing. **Earl Creech:** Conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; supervision; writing—original draft; writing—review and editing. **Burdette Barker:** Investigation; methodology; resources; software; validation; visualization; writing—original draft; writing—review and editing.

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